COMPUTATIONAL ANALYSIS AND SIMULATION OF TRANSIENT MOTION IN A CONSTRAINED SPRING-MASS SYSTEM

Aim

The aim of this project is to study and analyse the motion of a spring-mass system when a constraint is suddenly removed, and to investigate the conditions under which the object shows an initial upward movement instead of falling. This is done by examining how the forces acting on the system change due to the release of stored elastic energy in the springs, leading to a temporary imbalance of forces and resulting in transient oscillatory motion.

Introduction

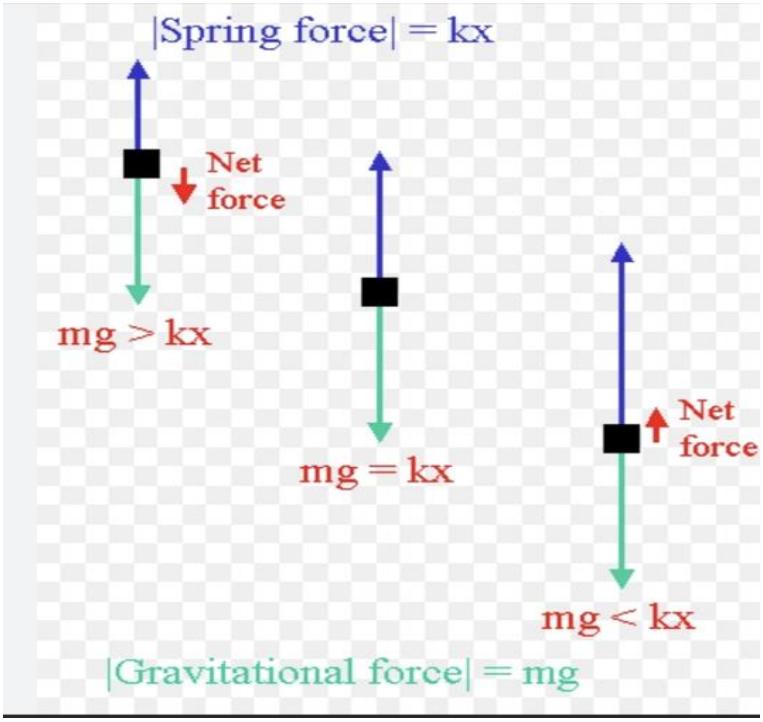
In general, when a supporting string of an object is cut, the object is expected to fall downward due to the action of gravity. However, under certain conditions, a system may exhibit unexpected behaviour. This project explores such a situation using a spring-mass arrangement in which the object initially moves upward instead of falling when a constraint is suddenly removed. The system consists of springs and a mass connected in a specific configuration that stores elastic energy when stretched. When the middle string is cut, the arrangement of the system changes instantly, leading to a temporary imbalance of forces. This causes the spring force to momentarily exceed the weight of the object, resulting in an upward motion followed by oscillations. This experiment aims to study and analyse this transient motion and understand the role of stored energy and force redistribution in determining the motion of the system.

Assumptions

- The springs used in the experiment are considered to be ideal, meaning their force increases proportionally with extension.
- The mass of the springs and strings is negligible compared to the mass of the body.
- Air resistance and frictional effects are neglected.
- The motion of the system is assumed to be vertical only, with no sideways movement.
- The system is initially in equilibrium before the middle string is cut.
- The cutting of the string is assumed to be instantaneous, causing a sudden change in the system.

- Energy losses due to heat or sound during motion are considered negligible.

Theory and Derivation of Formulae



Consider a mass attached to a spring and allowed to move vertically. The forces acting on the mass are:
Weight of the mass (mg), acting downward.
Spring force (F_{spring}), acting upward.
The net force acting on the mass is given by:
$F_{net} = F_{spring} - mg$
Initially, the system is at rest, so the net force is zero:
$F_{net} = 0$
Therefore:
$F_{spring} = mg$
Let x be the extension of the spring from its natural length. The spring force is directly proportional to this extension:
$F_{spring} \propto x$
Therefore:
$F_{spring} = kx$
Substituting in the net force equation:
$F_{net} = kx - mg$
For upward motion:
$kx > mg$
Thus, the mass moves upward when the spring force exceeds the gravitational force.

Description of the Experimental Setup:

1. The experimental setup consists of a vertical arrangement involving a rigid support, two springs, a light string with two knots, and a mass.

2. A spring is suspended from the rigid support at the top. The lower end of this spring is connected to a light string. This string contains two knots at different points along its length.

3. The upper knot of the string is attached to a second spring, whose other end is fixed to the rigid support. The lower end of the string is connected to a mass, which hangs freely under the action of gravity.

4. Thus, the system forms a combined spring–string arrangement, where both springs are initially stretched due to the weight of the mass, and the system is allowed to reach a state of equilibrium.

5. The portion of the string between the two knots acts as a connecting segment, which can be cut suddenly using scissors to change the configuration of the system and observe the resulting motion of the mass.

6. A vertical scale is placed alongside the setup to measure the displacement of the mass during the experiment.

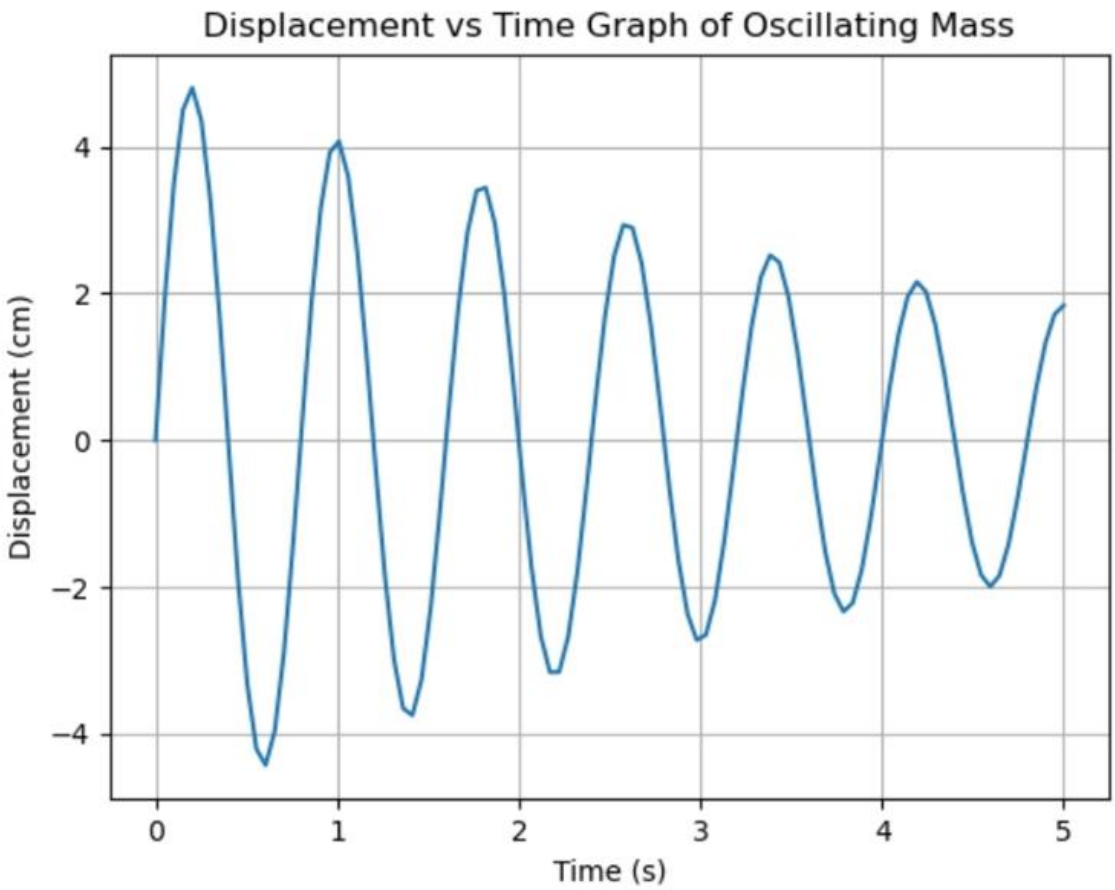
Procedure

- Arrange the experimental setup by suspending a spring from a rigid support. Attach a light string to the lower end of this spring and tie two knots at different points along the string. Connect the upper knot to a second spring fixed to the rigid support, and attach a mass to the lower end of the string.
- Ensure that all connections are secure and that the system is aligned vertically. Place a vertical scale beside the setup to measure the displacement of the mass.
- Allow the system to come to a state of rest so that both springs are stretched and the mass is in equilibrium.
- Note the initial position of the mass on the scale.
- Carefully cut the portion of the string between the two knots using scissors, ensuring that the cut is sudden and does not disturb the system externally.
- Observe the motion of the mass immediately after cutting. Record whether the mass initially moves upward and note the maximum upward displacement using the scale.
- Allow the mass to oscillate freely and observe the subsequent motion. Record additional readings if required for multiple trials.
- Repeat the experiment by resetting the setup to improve accuracy and consistency of observations.

Observation table:

Time(s)	displacement(cm)
0	0
0.5	-4.3
1	4
1.5	-3.7
2	3.4
2.5	-3.1
3	2.8
3.5	-2.5
4	2.2
4.5	-2
5	1.8

Graph(displacement vs time):



Analysis of Observations:

- From the observations recorded, it is evident that immediately after the middle portion of the string is cut, the mass initially moves upward instead of falling downward. This indicates that the upward force acting on the mass at that instant is greater than its weight. This can be explained by the fact that both springs are initially stretched and store elastic potential energy. When the constraint is suddenly removed, this stored energy is released, producing a net upward force on the mass.

- The recorded values of maximum upward displacement show slight variations across trials, which may be due to minor disturbances during cutting or measurement errors. However, the overall trend remains consistent, confirming the repeatability of the result.

- The time period of oscillation is observed to be nearly constant for all trials, indicating that the system undergoes periodic motion. After the initial upward movement, the mass oscillates about a new equilibrium position, gradually decreasing in amplitude due to energy loss in the form of air resistance and internal friction in the springs (damping effect).

- Thus, the observations support the theoretical condition that when the spring force exceeds the gravitational force ($kx > mg$), the mass experiences an upward motion, followed by oscillatory behaviour.

Explanation of the Observed Motion:

- When the system is initially at rest, the forces acting on the mass are balanced, with the upward spring force equal to the downward weight of the mass. At this stage, both springs are stretched and store elastic potential energy.

- When the middle portion of the string is suddenly cut, the arrangement of the system changes instantly. However, the springs do not immediately change their length, so they remain stretched for a brief moment. Due to this, the spring force acting on the mass remains large, while the constraints on the system are reduced.

- As a result, the upward spring force becomes greater than the gravitational force acting downward. This creates a net upward force on the mass, causing it to move upward initially instead of falling.

- After this initial upward motion, the mass does not continue moving upward indefinitely. Instead, it begins to oscillate about a new equilibrium position. This oscillatory motion occurs because the spring force and gravitational force continuously try to restore balance in the system. Over time, the amplitude of oscillation gradually decreases due to energy loss caused by air resistance and internal friction in the springs.

- Thus, the observed motion can be explained by the sudden release of stored elastic energy and the temporary imbalance of forces in the system.

Conclusion:

This experiment demonstrates that a mass attached to a spring system can exhibit an initial upward motion when a constraint is suddenly removed. Although the system is initially in equilibrium, the springs store elastic potential energy due to their extension. When the middle portion of the string is cut, the springs remain stretched momentarily, resulting in an unbalanced force where the spring force exceeds the weight of the mass. This causes the mass to move upward. The system then undergoes oscillatory motion about a new equilibrium position, with the amplitude gradually decreasing due to damping effects. Thus, the observations are consistent with the condition that upward motion occurs when the spring force is greater than the gravitational force.

Applications:

- Mechanical suspension systems (shock absorbers)
- Spring-based lifting and release mechanisms
- Vibration control in engineering structures
- Design of safety devices using stored elastic energy
- Analysis of oscillatory systems in physics and engineering

Limitations:

- Air resistance affects motion and reduces accuracy
- Friction within springs causes energy loss (damping)
- Difficulty in achieving a perfectly sudden cut
- Measurement errors in displacement and time
- Springs may not behave ideally under all conditions

Future Improvements:

- Use springs with known and precise spring constants
- Implement a controlled cutting mechanism for consistency
- Record motion using high-speed video for accurate analysis
- Use digital sensors to measure displacement and time
- Reduce external disturbances to improve reliability